

Module 02: Agents — Neural Correlates of the Self-Model

Course: Active Inference 401 | **Unit:** Neuroscientific Frontiers | **Audience:**
Advanced undergraduates / graduate students

Learning Objectives

1. Identify the **neural substrates** of the self-model — interoceptive, proprioceptive, and narrative self-representations.
2. Analyze the **default mode network (DMN)** as the neural basis of the generative model of self.
3. Evaluate how **interoceptive inference** grounds subjective experience and emotional awareness.
4. Connect self-model neuroscience to the Philosophical Foundations treatment of selfhood as inference (Track 1, Module 02).

Introduction

If Active Inference treats every agent as a generative model, then selfhood is a specific kind of prediction — the model the brain constructs about its own states. This module maps the "self as model" onto specific neural circuits.

Key Insight — The Self is a Prediction: The self is not a fixed entity residing somewhere in the brain. It is a continuously updated predictive model — the brain's best hypothesis about who it is, what it feels, and where it stands in the world. Self-model disruptions produce some of the most dramatic clinical phenomena in neurology and psychiatry.

Key Concepts

1. The Default Mode Network as Self-Model

The DMN — a network of brain regions active during rest, mind-wandering, and self-referential thought — has been proposed as the neural substrate of the self-model in Active Inference:

Key DMN regions and their self-model functions:

- **Medial prefrontal cortex (mPFC):** Self-referential processing — "what am I thinking/feeling?" Encodes higher-order predictions about one's own mental states.
- **Posterior cingulate cortex (PCC):** Autobiographical memory and narrative self — "who am I over time?" Integrates past experiences into a coherent self-narrative.
- **Angular gyrus / temporoparietal junction (TPJ):** Self-other distinction and perspective-taking — "how do others see me?" The interface between self-model and social inference.
- **Hippocampal formation:** Episodic memory and temporal self — "where am I in my life story?" Provides the temporal scaffolding for narrative identity.
- **Precuneus:** Visuospatial self-representation and first-person perspective — "where am I in space?" Implements the spatial component of the embodied self.

The DMN implements the **highest levels of the generative model** — the most abstract, temporally extended predictions about oneself and one's situation. When externally directed attention is not required, the DMN generates predictions about the self's past, present, and future states.

Cross-Track Connection — Philosophical Foundations (Module 02): The philosophical distinction between the "minimal self" (pre-reflective bodily awareness) and the "narrative self" (autobiographical identity) maps onto the neural distinction between interoceptive/proprioceptive circuits (insular cortex, somatosensory cortex) and the DMN (mPFC, PCC, hippocampus). Philosophy identified the distinction; neuroscience localizes it.

2. Interoceptive Inference and the Embodied Self

Seth & Friston (2016) propose that **interoceptive inference** — the brain's predictions about the body's internal states — is the foundation of subjective experience:

The interoceptive hierarchy:

- **Peripheral level:** Visceral receptors detect heart rate, respiration, gut motility, blood chemistry, temperature
- **Brainstem level:** Nucleus of the solitary tract (NTS) and parabrachial nucleus process raw visceral signals
- **Insular cortex:** The primary interoceptive cortex — builds a map of the body's internal state. The posterior insula processes raw interoceptive signals; middle insula integrates these with contextual information
- **Anterior insula:** Integrates interoceptive signals with emotional, cognitive, and social context — the "sentient self." The anterior insula is consistently activated during subjective emotional experience across all emotion categories

The key insight: Emotions are interoceptive prediction errors. When you feel anxious, your brain is predicting a deviation from expected bodily states (elevated heart rate, shallow breathing). The *feeling* of anxiety is the interoceptive prediction error itself — the discrepancy between predicted and actual bodily states.

Clinical Vignette — Heartbeat Evoked Potentials: The heartbeat evoked potential (HEP) — a cortical response time-locked to each heartbeat — is larger in individuals with better interoceptive accuracy and correlates with anterior insular thickness. Patients with alexithymia (difficulty identifying emotions) show reduced HEPs, supporting the link between interoceptive precision and emotional awareness.

3. Proprioceptive Inference and Body Ownership

The sense of owning and controlling one's body depends on proprioceptive inference:

Body schema: A dynamic generative model of the body's position, posture, and boundaries. This model is maintained by continuous proprioceptive prediction and correction. The

posterior parietal cortex maintains the body schema; the premotor cortex generates predictions about upcoming body states.

The rubber hand illusion: When visual and tactile information is artificially correlated with a fake hand, the proprioceptive generative model updates to "incorporate" the fake hand — prediction errors are minimized by shifting the body boundary. Brain imaging shows this illusion is associated with changes in premotor and intraparietal cortex, and a decrease in proprioceptive precision over the real hand.

Phantom limbs: After amputation, the proprioceptive model persists, generating predictions about a limb that no longer exists. The resulting prediction errors (which cannot be resolved) produce phantom sensations. Mirror therapy works by providing visual prediction error that can recalibrate the phantom limb model.

Full-body illusions: Virtual reality induction of full-body ownership transfer demonstrates that the body model is remarkably plastic. The multisensory integration hypothesis (Blanke, 2012) maps onto Active Inference: body ownership reflects the posterior probability that "this body is mine" given multisensory evidence.

4. Neural Markers of Self-Model Disruption

Disruptions to the self-model produce characteristic clinical phenomena:

Condition	Neural Basis	Self-Model Disruption	Active Inference Mechanism
Depersonalization	Reduced insular and DMN activity	Weakened interoceptive self-model; "I don't feel real"	Low precision on interoceptive prediction errors
Out-of-body experience	TPJ disruption	Disrupted proprioceptive self-localization	Conflict between vestibular and visual self-model
Anosognosia	Right parietal damage	Inability to update self-model to include deficit	Fixed prior that "I am able" resists error-driven updating
Dissociative identity	Altered DMN functional connectivity	Fragmented self-model with multiple modes	Multiple competing generative models of self
Alien hand syndrome	SMA lesion	Action without self-attribution	Motor model disconnected from self-model
Cotard's syndrome	Insular/DMN disconnection	Belief that one is dead	Catastrophic interoceptive prediction error — no "aliveness" signal
Somatoparaphrenia	Right parietal damage	Denial of limb ownership	Proprioceptive model excludes the limb

Key Insight — The Self as Hypothesis, Not Axiom: Notice that the self-model is not inviolable — it can fragment, shift, expand, shrink, or even conclude that the host organism is dead (Cotard's). The self is the brain's "best guess" about its

own identity, and like all guesses, it can be systematically wrong. This has profound implications for understanding both psychopathology and contemplative practices (meditation, which may systematically reduce self-model precision).

5. The Minimal Self and Neural Correlates of Consciousness

The "minimal self" (Zahavi, 2005) — the pre-reflective sense of being a subject of experience — may correspond to the most basic level of interoceptive and proprioceptive inference:

Hypothesis: Consciousness is present whenever a system maintains an active self-model — a predictive model of its own internal and bodily states. NCC (neural correlates of consciousness) should therefore track interoceptive and proprioceptive prediction circuits rather than specific content (faces, colors, etc.).

Evidence: Anesthesia disproportionately disrupts interoceptive and DMN processing; loss of consciousness correlates with loss of self-referential prediction rather than loss of sensory processing. The posterior "hot zone" (posterior cingulate, precuneus, lateral temporal-parietal cortex) identified by Koch et al. (2016) overlaps significantly with the self-modeling network.

The dreaming puzzle: During REM sleep, the self-model is partially maintained (you experience yourself as an agent in the dream) but decoupled from sensory prediction. This dissociation supports the view that consciousness tracks the self-model rather than sensory processing per se.

6. Development of the Self-Model

The self-model is not innate but develops through experience:

Neonatal period: Basic interoceptive self — distinguishing self-generated from external touch (based on predicted sensory consequences of motor actions). Newborns already show differential responses to self-touch vs. external touch.

First year: Body schema development — integrating proprioceptive, vestibular, and visual information into a coherent body model. The mirror self-recognition test (passing around 18-24 months) marks a milestone in higher-order self-modeling.

Childhood-adolescence: Narrative self-construction — building autobiographical memory and integrating social feedback into the self-model. DMN functional connectivity matures throughout adolescence.

Summary

The self-model is neurally implemented across the DMN (narrative self), insular cortex (interoceptive self), and proprioceptive circuits (bodily self). Emotions arise as interoceptive prediction errors. Clinical disruptions of the self-model (depersonalization, phantom limbs, out-of-body experiences, Cotard's syndrome) confirm that selfhood is an inference that can systematically fail. Consciousness may track the integrity of the self-model. The self-model develops through experience from basic interoceptive differentiation to full narrative identity.

Further Reading

- Seth, A. K. (2013). Interoceptive inference, emotion and the embodied self. *Trends in Cognitive Sciences*, 17(11), 565-573.
- Blanke, O. (2012). Multisensory brain mechanisms of bodily self-consciousness. *Nature Reviews Neuroscience*, 13(8), 556-571.
- Andrews-Hanna, J. R. et al. (2014). The default network and self-generated thought. *Annals NY Acad Sciences*, 1316, 29-52.
- Koch, C. et al. (2016). Neural correlates of consciousness: progress and problems. *Nature Reviews Neuroscience*, 17(5), 307-321.
- Tsakiris, M. (2010). My body in the brain: A neurocognitive model of body-ownership. *Neuropsychologia*, 48(3), 703-712.